

Embedded Linux & System Software Interview Handbook

Chapter 2: Complete Linux Boot Process

Goal: Understand every stage from pressing the power button until the Linux login prompt appears.

Overall Flow

Power On → BootROM → SPL (optional) → U-Boot → Linux Kernel → Device Tree →
initramfs/initrd → Root File System → systemd → Login Prompt

1. Power On

The CPU starts execution from a fixed reset vector. Registers are initialized and execution jumps into the immutable BootROM present inside the SoC.

2. BootROM

BootROM selects the configured boot device (eMMC, NAND, NOR Flash, SD Card or Network), performs minimal hardware initialization and loads the first-stage bootloader (often SPL). In secure boot systems it may also verify signatures.

3. SPL (Secondary Program Loader)

SPL is a tiny bootloader used when full U-Boot cannot fit in internal SRAM. Its main responsibility is DDR initialization so that larger software can be loaded into external RAM.

4. U-Boot

U-Boot initializes clocks, DDR, console (UART), storage, Ethernet and other peripherals. It loads the Linux kernel image, Device Tree Blob (DTB) and initramfs into RAM, prepares kernel boot arguments and transfers control to the kernel.

Useful U-Boot Commands

printenv, setenv, saveenv, booti, bootm, ext4load, fatload, tftpboot.

5. Linux Kernel

The kernel decompresses itself, initializes memory management, scheduler, interrupt controller, timers and device drivers. It parses the Device Tree to discover hardware and mounts the initial root filesystem.

6. Device Tree (DTB)

The Device Tree describes CPU, memory, UART, I2C, SPI, GPIO, Ethernet and interrupts. It allows one kernel image to support multiple hardware boards without recompilation.

7. initramfs / initrd

A temporary root filesystem containing essential drivers and initialization scripts. It prepares storage, loads required modules and eventually switches to the real root filesystem.

8. Root File System

The kernel mounts the real root filesystem (eMMC, SSD, NFS etc.) and starts the first user-space process, usually systemd.

9. systemd

systemd starts services, networking, logging, login manager and application processes according to dependencies and target configuration.

Debugging the Boot Process

Use UART to capture early boot logs. In U-Boot use 'printenv' to inspect boot parameters. In Linux use dmesg, journalctl -b and /proc/cmdline. If the board hangs, identify whether the last log belongs to BootROM, U-Boot, kernel or user space.

Real Samsung Board Bring-up Scenario

During initial board bring-up, UART output is often the only available visibility. Engineers verify that DDR initialization succeeds, U-Boot starts correctly, the kernel identifies memory, the Device Tree

matches the hardware and required drivers probe successfully.

Common Interview Questions

1. Explain the Linux boot process.
2. What is BootROM?
3. Why is SPL required?
4. What is U-Boot?
5. Difference between U-Boot and BIOS/UEFI?
6. What is a Device Tree?
7. What is initramfs?
8. How would you debug a board that hangs during boot?
9. What information is passed through bootargs?
10. How does the kernel discover hardware?

Key Takeaways

Know the responsibility of every stage, the handoff between components and how to identify failures using UART logs. This chapter forms the foundation for Board Bring-up and BSP.